

# 1.Green field scenario

The screenshot shows the Visual Studio Code interface with the Explorer, Problems, Output, Debug Console, and Terminal panes. The Explorer pane on the left shows a project structure with folders like `01-greenfield-interactive`, `02-brownfield-interactive`, `03-scripted-interactive`, `04-greenfield-runtime`, `05-brownfield-runtime`, `06-ambiguous-runtime`, `07-governance-failure`, `08-reliability`, `09-tests`, `10-architecture`, `11-demo-video`, `generated_artifacts`, `migrations`, `prompts`, `scripts`, `tests`, `workspace`, `env`, `env.example`, `.gitignore`, `alembic.ini`, `CODEX_CONTEXT_HANDOFF.md`, `CODEX_VERIFICATION_STATUS.md`, `envobject.toml`, `OUTLINE`, and `TIMELINE`. The Problems pane shows a list of errors. The Output pane shows the output of the `Start-Transcript` command. The Debug Console pane shows the output of the `Start-Transcript` command. The Terminal pane shows the output of the `Start-Transcript` command.

```
(.venv) PS C:\Code\agentic-url-shortner> Start-Transcript `
>> -Path evidence\01-greenfield-interactive\greenfield-transcript.txt `
>> -Force
Transcript started, output file is evidence\01-greenfield-interactive\greenfield-transcript.txt
(.venv) PS C:\Code\agentic-url-shortner>
(.venv) PS C:\Code\agentic-url-shortner> .\scripts\run_phased_scripted_demo.ps1 `
>> -Mode Interactive `
>> -Scenario Greenfield `
>> -WorkspacePath "phased-interactive"
Deterministic scripted URL-shortener demonstration.
LUN responses are scripted; orchestration, approvals, editing, Git, validation, retries, rollback, and audit execution remain real.
Expected provider mode: SCRIPTED
Starting Uvicorn on host 127.0.0.1, port 8000...
Server is live and ready at http://127.0.0.1:8000.

Creating GREENFIELD workflow in workspace 'phased-interactive-greenfield-20260803182759430'...
[Greenfield] status-WAITING_FOR_REQUIREMENT_APPROVAL stage-REQUIREMENT_APPROVAL stateVersion-2

Approval gate observed
Scenario: Greenfield
Workflow ID: MF-7989E1418E49
Current stage: REQUIREMENT_APPROVAL
Gate type: REQUIREMENT
Approval ID: APR-87777B1722E4
State version: 2
Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
Architecture version: 0
Plan version: 0
Retry counts: {}
Available decisions:
1. APPROVE
2. APPROVE_WITH_CONDITIONS
3. REQUEST_CHANGES
4. REJECT
Select an allowed action: 1
Optional approval comments: Requirements Approved
[Greenfield] status-WAITING_FOR_ARCHITECTURE_APPROVAL stage-ARCHITECTURE_APPROVAL stateVersion-3

Approval gate observed
Scenario: Greenfield
Workflow ID: MF-7989E1418E49
Current stage: ARCHITECTURE_APPROVAL
Gate type: ARCHITECTURE_AND_PLAN
Approval ID: APR-DE783030A9E
State version: 3
Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
Architecture version: 1
Plan version: 1
Retry counts: {}
Available decisions:
1. APPROVE
2. APPROVE_WITH_CONDITIONS
3. REQUEST_CHANGES
4. REJECT
Select an allowed action: 1
```

The screenshot shows the Visual Studio Code interface with the Explorer, Problems, Output, Debug Console, and Terminal panes. The Explorer pane on the left shows a project structure with folders like `01-greenfield-interactive`, `02-brownfield-interactive`, `03-scripted-interactive`, `04-greenfield-runtime`, `05-brownfield-runtime`, `06-ambiguous-runtime`, `07-governance-failure`, `08-reliability`, `09-tests`, `10-architecture`, `11-demo-video`, `generated_artifacts`, `migrations`, `prompts`, `scripts`, `tests`, `workspace`, `env`, `env.example`, `.gitignore`, `alembic.ini`, `CODEX_CONTEXT_HANDOFF.md`, `CODEX_VERIFICATION_STATUS.md`, `envobject.toml`, `OUTLINE`, and `TIMELINE`. The Problems pane shows a list of errors. The Output pane shows the output of the `Start-Transcript` command. The Debug Console pane shows the output of the `Start-Transcript` command. The Terminal pane shows the output of the `Start-Transcript` command.

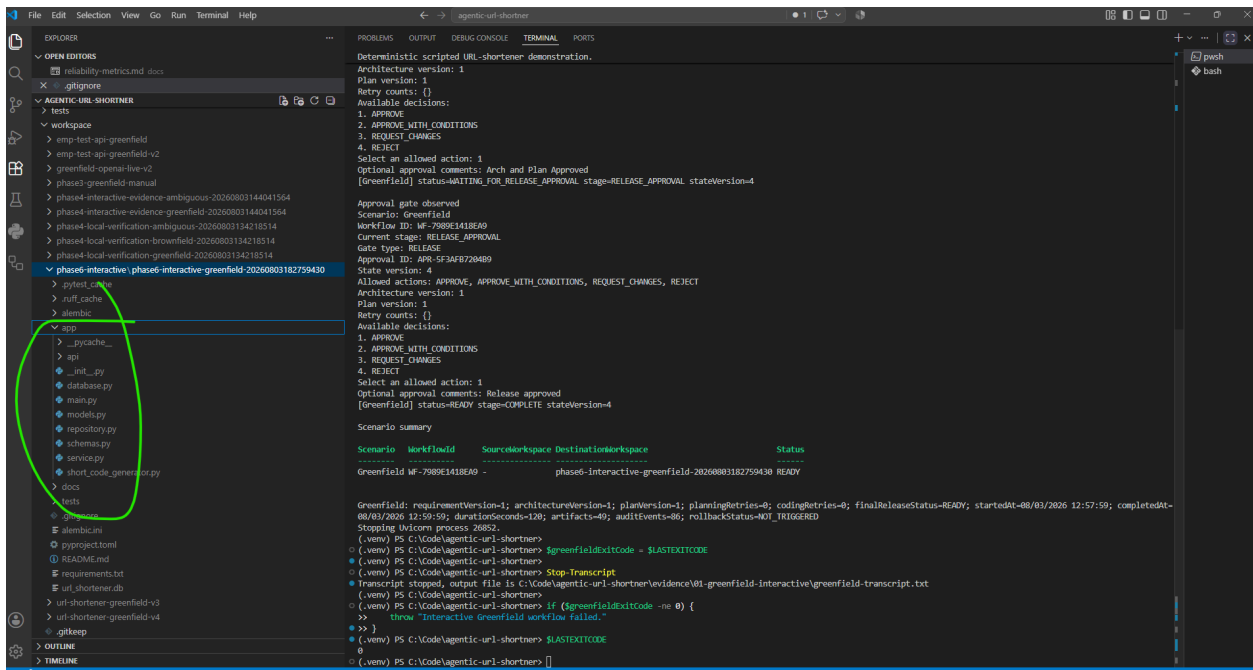
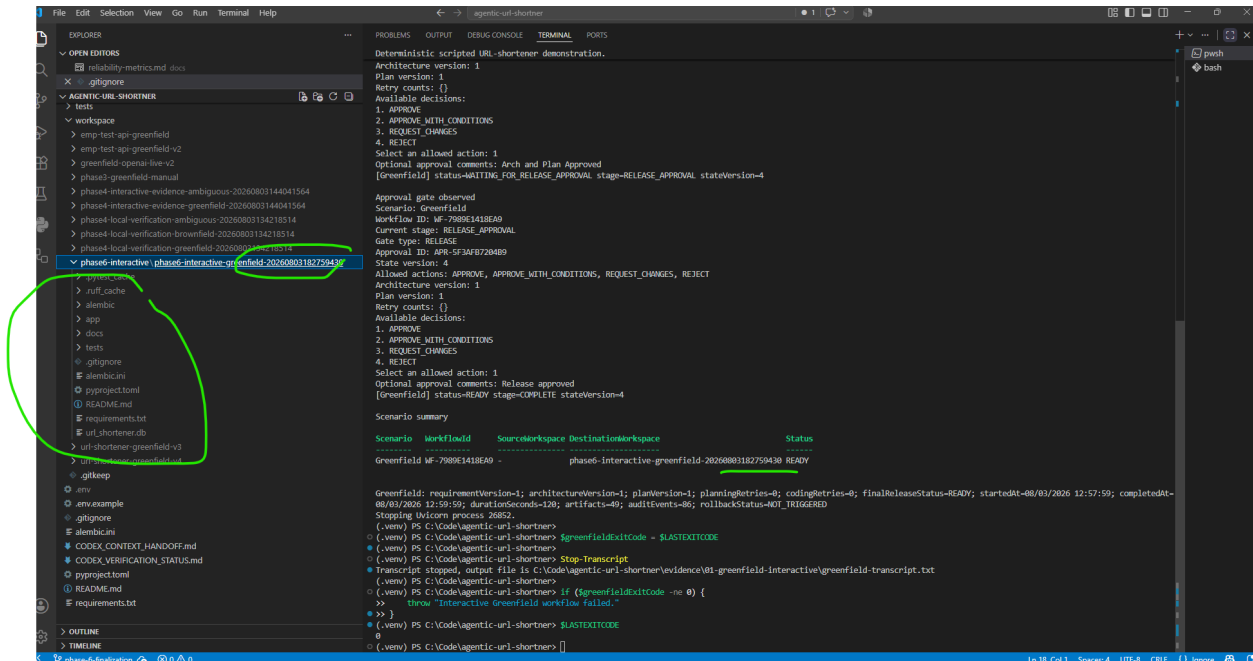
```
(.venv) PS C:\Code\agentic-url-shortner> Start-Transcript `
>> -Path evidence\01-greenfield-interactive\greenfield-transcript.txt `
>> -Force
Transcript started, output file is evidence\01-greenfield-interactive\greenfield-transcript.txt
(.venv) PS C:\Code\agentic-url-shortner>
(.venv) PS C:\Code\agentic-url-shortner> .\scripts\run_phased_scripted_demo.ps1 `
>> -Mode Interactive `
>> -Scenario Greenfield `
>> -WorkspacePath "phased-interactive"
Deterministic scripted URL-shortener demonstration.
LUN responses are scripted; orchestration, approvals, editing, Git, validation, retries, rollback, and audit execution remain real.
Expected provider mode: SCRIPTED
Starting Uvicorn on host 127.0.0.1, port 8000...
Server is live and ready at http://127.0.0.1:8000.

Creating GREENFIELD workflow in workspace 'phased-interactive-greenfield-20260803182759430'...
[Greenfield] status-WAITING_FOR_REQUIREMENT_APPROVAL stage-REQUIREMENT_APPROVAL stateVersion-2

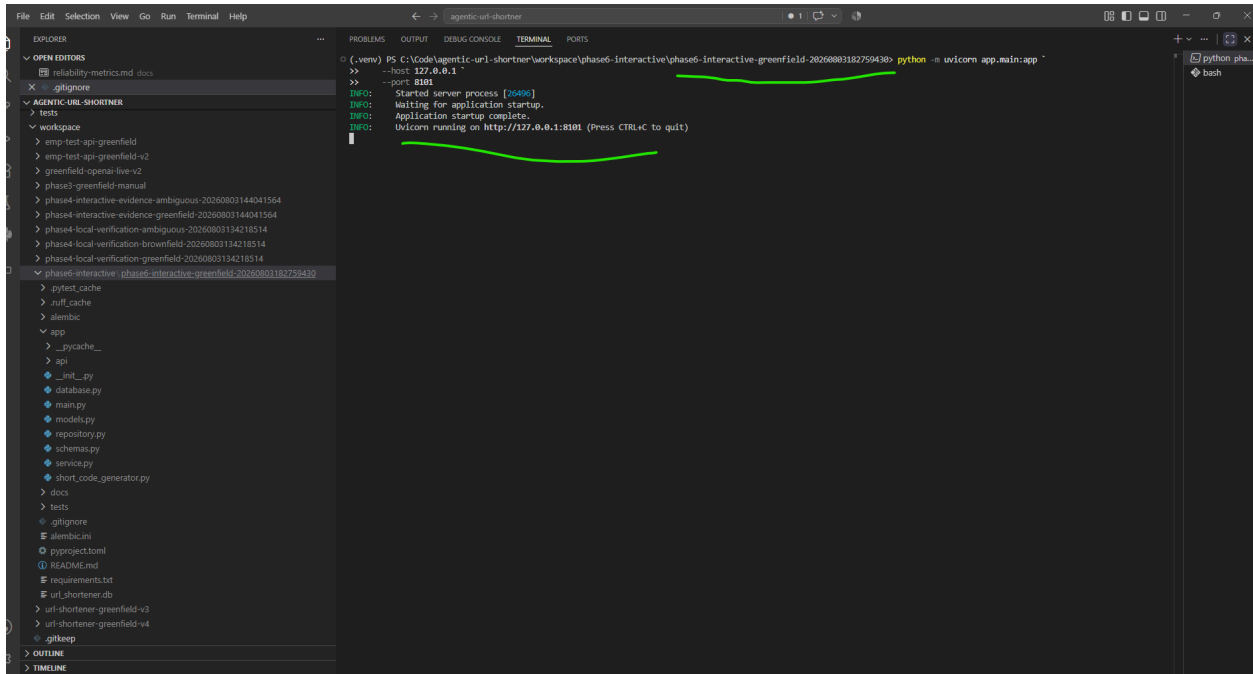
Approval gate observed
Scenario: Greenfield
Workflow ID: MF-7989E1418E49
Current stage: REQUIREMENT_APPROVAL
Gate type: REQUIREMENT
Approval ID: APR-87777B1722E4
State version: 2
Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
Architecture version: 0
Plan version: 0
Retry counts: {}
Available decisions:
1. APPROVE
2. APPROVE_WITH_CONDITIONS
3. REQUEST_CHANGES
4. REJECT
Select an allowed action: 1
Optional approval comments: Requirements Approved
[Greenfield] status-WAITING_FOR_ARCHITECTURE_APPROVAL stage-ARCHITECTURE_APPROVAL stateVersion-3

Approval gate observed
Scenario: Greenfield
Workflow ID: MF-7989E1418E49
Current stage: ARCHITECTURE_APPROVAL
Gate type: ARCHITECTURE_AND_PLAN
Approval ID: APR-DE783030A9E
State version: 3
Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
Architecture version: 1
Plan version: 1
Retry counts: {}
Available decisions:
1. APPROVE
2. APPROVE_WITH_CONDITIONS
3. REQUEST_CHANGES
4. REJECT
Select an allowed action: 1
Optional approval comments: Arch and Plan Approved
```





Generated App URL shortener is running



## URL Shortener 0.1.0 QAS 3.0

/openapi.json

### default

|      |                 |            |
|------|-----------------|------------|
| POST | /api/v1/urls    | Create Url |
| GET  | /({short_code}) | Redirect   |

### Schemas

HTTPValidationError > Expand all object

UrlCreate > Expand all object

UrlResponse > Expand all object

ValidationError > Expand all object

URL Shortener - Swagger UI

127.0.0.1:8101/docs#/default/create\_url\_api\_v1\_urls\_post

Edit Value Schema

```
{
  "original_url": "https://www.schwab.com/invest-with-us/self-directed-investing"
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8101/api/v1/urls' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "original_url": "https://www.schwab.com/invest-with-us/self-directed-investing"
  }'
```

Request URL

http://127.0.0.1:8101/api/v1/urls

Server response

Code Details

201

Response body

```
{
  "original_url": "https://www.schwab.com/invest-with-us/self-directed-investing",
  "short_code": "kk718bh",
  "short_url": "http://127.0.0.1:8101/kk718bh"
}
```

Response headers

```
content-length: 149
content-type: application/json
date: Mon, 05 Aug 2024 15:15:01 GMT
```

New InPrivate tab

http://127.0.0.1:8101/kk718bh

InPrivate browsing

InPrivate search with Microsoft Bing

What InPrivate browsing does

- Deletes your browsing info when you close all InPrivate windows
- Saves collections, favorites, and downloads (but not download history)
- Prevents Microsoft Bing searches from being associated with you

What InPrivate browsing doesn't do

- Hide your browsing from your school, employer, or internet service provider
- Give you additional protection from tracking by default
- Add additional protection to what's available in normal browsing

Always use "Strict" tracking prevention when browsing InPrivate

If this is off, we'll use the same tracking prevention setting as a normal browsing window

Cookies aren't shared during InPrivate

InPrivate browsing keeps cookies private to help prevent sites from sharing your data across the web. If a site doesn't work, you can temporarily allow cookie access.

More details

The top image shows the Charles Schwab website's 'Self-directed investing' page. It features a navigation bar with links like 'Accounts & Products', 'Advice', 'Pricing', 'Why Schwab', and 'Learn'. The main heading is 'Self-directed investing' with a subtext: 'Many self-directed investors want research depth and tools—more than the lowest fee. Schwab can help deliver both.' Below this is an 'Open an account' button. To the right, there are screenshots of the Schwab platform's 'Portfolio Builder' and 'Thematic Stock Sets' features.

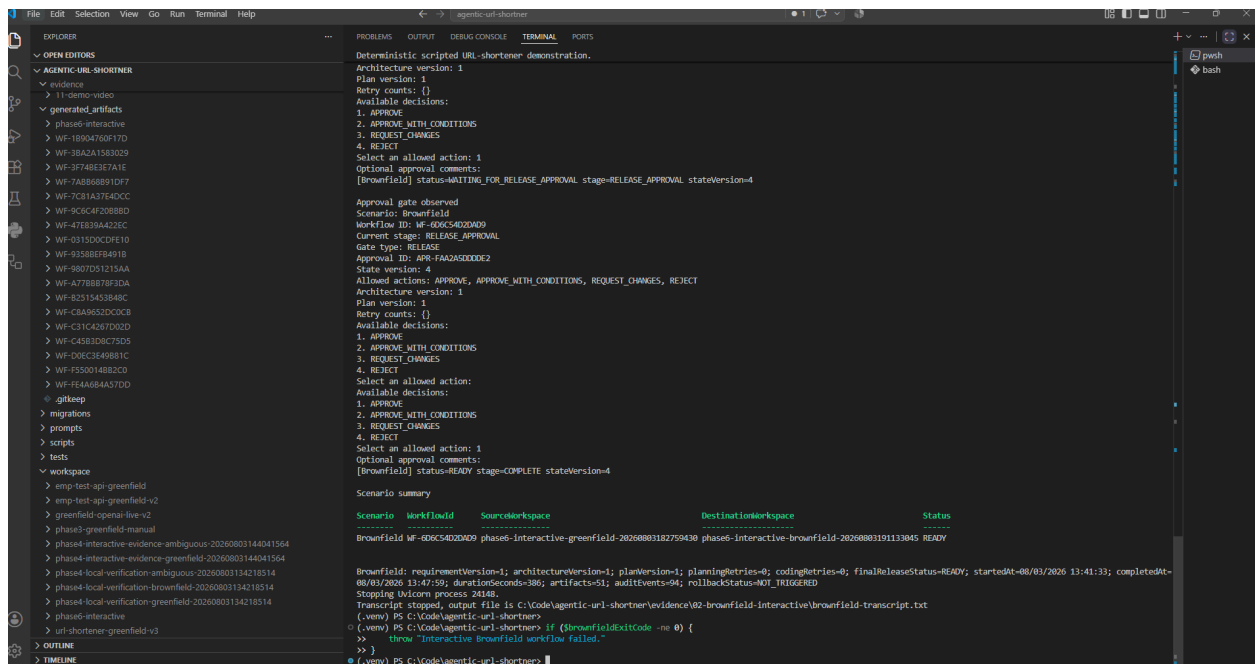
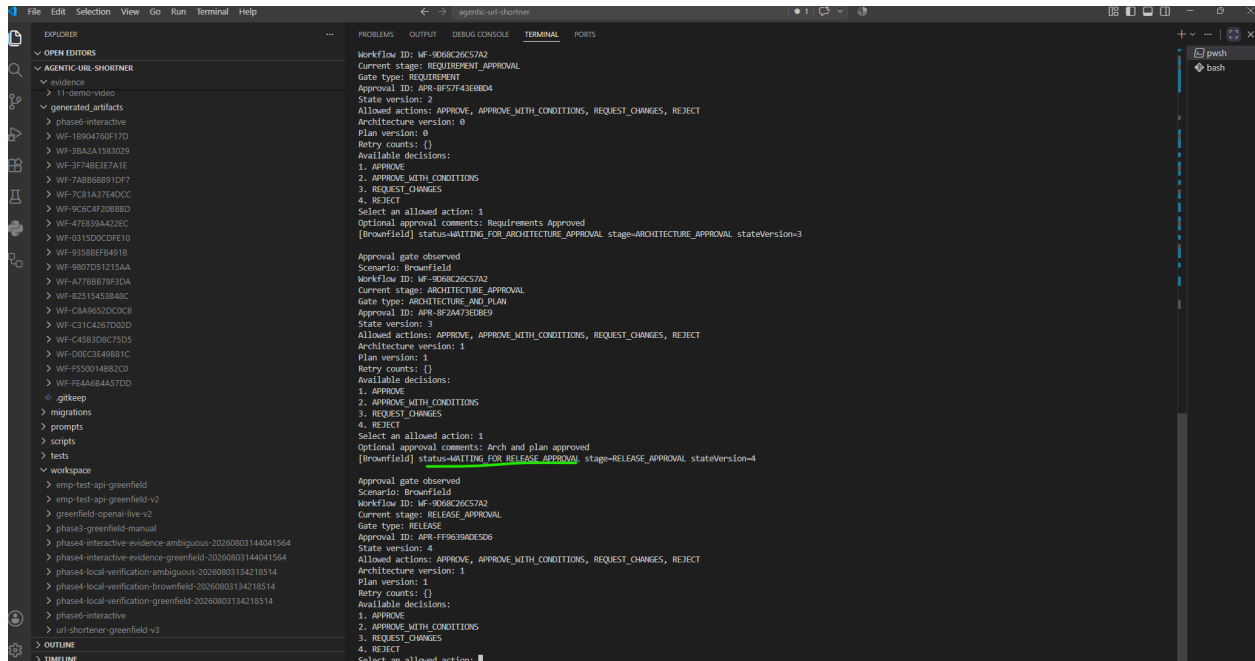
The bottom image is a StockBrokers.com award graphic for 2026. It includes a 'Cookie Consent' pop-up on the left. The main text reads 'Full-Service Broker by StockBrokers.com 2026'. Below this, there are three award categories, each with a 'Best In Class' badge: '#1', 'Best In Class', and 'Best In Class'.

## 2. Brownfield Scenario

```

File Edit Selection View Go Run Terminal Help
EXPLORER
  OPEN EDITORS
  AGENTIC-URL-SHORTENER
    evidence
    11-demo-video
    generated_artifacts
    phases-interactive
      WF-1B50470F97D
      WF-3BA2A15A93D29
      WF-3748E3E7A3E
      WF-7AB866B91D07
      WF-7C81A37E4DCC
      WF-9C8C4F20B8BD
      WF-47E839A422EC
      WF-0315D0CDFE10
      WF-93588EF4918
      WF-9807051215AA
      WF-A7789B791D0A
      WF-82215433B48C
      WF-C8A9652D0C0B
      WF-C31C4267D0D0
      WF-C458108C7305
      WF-D0EC3E498B1C
      WF-F350014B82C0
      WF-FEA68A457D0
    .gitkeep
    migrations
    prompts
    scripts
    tests
    workspace
      emp-test-api-greenfield
      emp-test-api-greenfield-v2
      greenfield-openai-live-v2
      phase3-greenfield-manual
      phase4-interactive-evidence-ambiguous-20260803144041564
      phase4-local-verification-greenfield-20260803134218514
      phase4-local-verification-greenfield-20260803134218514
      phase4-local-verification-greenfield-20260803134218514
      phase5-interactive
      url-shortener-greenfield-v3
    OUTLINE
    TIMELINE

TERMINAL
  Scenario Brownfield
  -SourceWorkspace $greenfieldSourceName
  -WorkspacePrefix "phase5-interactive"
  $BrownfieldExitCode = $LASTEXITCODE
  }
  } finally {
  Stop-Transcript
  }
  }
  Deterministic scripted URL-shortener demonstration.
  URI responses are scripted; orchestration, approvals, editing, Git, validation, retries, rollback, and audit execution remain real.
  Expected provider mode: SCRIPTED
  Starting Unicorn on host 127.0.0.1, port 8000...
  Server is live and ready at http://127.0.0.1:8000/
  Creating BROWNFIELD workflow in workspace "phase5-interactive-brownfield-20260803198615989"...
  [Brownfield] status=AWAITING_FOR_REQUIREMENT_APPROVAL stage=REQUIREMENT_APPROVAL stateVersion=2
  Approval gate observed
  Scenario: Brownfield
  Workflow ID: WF-S00826CS7A2
  Current stage: REQUIREMENT_APPROVAL
  Gate type: REQUIREMENT
  Approval ID: APR-0F57F43E0BD4
  State version: 2
  Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
  Architecture version: 0
  Plan version: 0
  Retry counts: {}
  Available decisions:
  1. APPROVE
  2. APPROVE_WITH_CONDITIONS
  3. REQUEST_CHANGES
  4. REJECT
  Select an allowed action: 1
  Optional approval comments: Requirements Approved
  [Brownfield] status=AWAITING_FOR_ARCHITECTURE_APPROVAL stage=ARCHITECTURE_APPROVAL stateVersion=3
  Approval gate observed
  Scenario: Brownfield
  Workflow ID: WF-S00826CS7A2
  Current stage: ARCHITECTURE_APPROVAL
  Gate type: ARCHITECTURE_AND_PLAN
  Approval ID: APR-0F3A473ED0E9
  State version: 3
  Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
  Architecture version: 1
  Plan version: 1
  Retry counts: {}
  Available decisions:
  1. APPROVE
  2. APPROVE_WITH_CONDITIONS
  3. REQUEST_CHANGES
  4. REJECT
  Select an allowed action:
  
```



Brownfield app running

URL Shortener - Swagger UI

127.0.0.1:8102/docs/#/default/create\_url\_api\_v1\_urls\_post

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8102/api/v1/urls' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "original_url": "https://www.schab.com/invest-with-us/self-directed-investing"
  }'
```

Request URL

http://127.0.0.1:8102/api/v1/urls

Server response

Code Details

201

Response body

```
{
  "original_url": "https://www.schab.com/invest-with-us/self-directed-investing",
  "short_code": "isafeswt",
  "short_url": "http://127.0.0.1:8102/isafeswt"
}
```

Response headers

```
content-length: 149
content-type: application/json
date: Mon, 03 Aug 2026 15:50:23 GMT
server: unknown
```

Responses

| Code | Description         | Links    |
|------|---------------------|----------|
| 201  | Successful Response | No links |

Media type

URL Shortener - Swagger UI

New InPrivate tab

http://127.0.0.1:8102/isafeswt

http://127.0.0.1:8102/isafeswt - Search

InPrivate browsing

InPrivate search with Microsoft Bing

What InPrivate browsing does

- Deletes your browsing info when you close all InPrivate windows
- Saves collections, favorites, and downloads (but not download history)
- Prevents Microsoft Bing searches from being associated with you

What InPrivate browsing doesn't do

- Hide your browsing from your school, employer, or internet service provider
- Give you additional protection from tracking by default
- Add additional protection to what's available in normal browsing

Always use "Strict" tracking prevention when browsing InPrivate

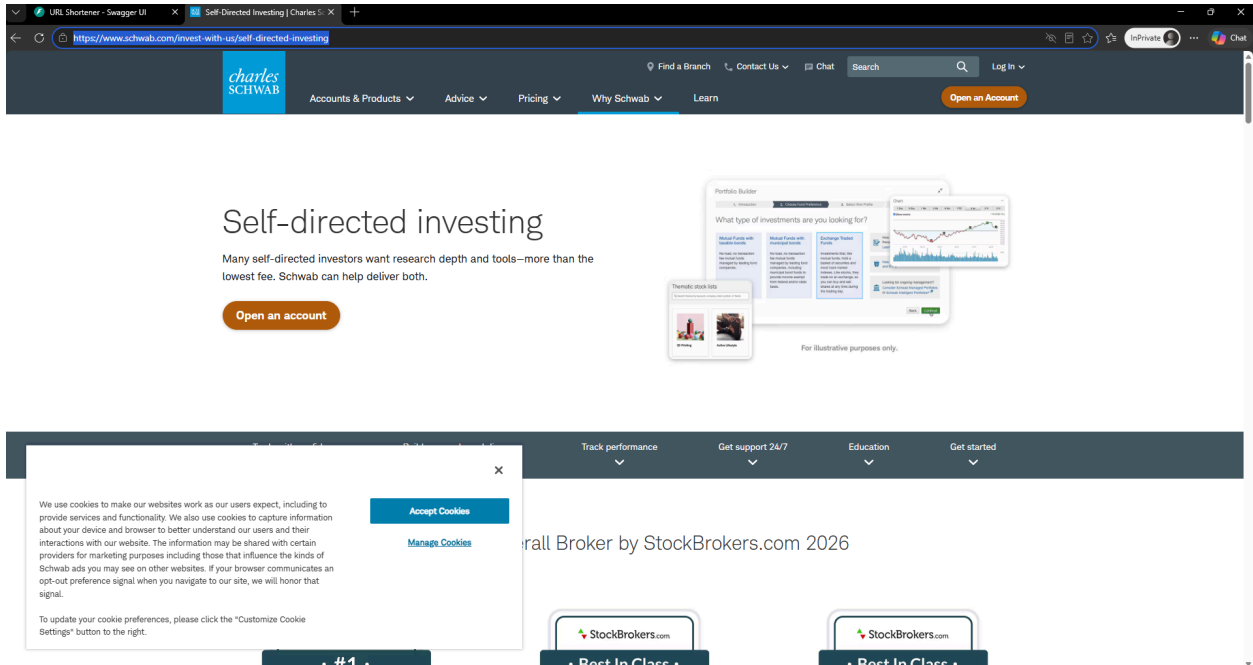
If this is off, we'll use the same tracking prevention setting as a normal browsing window

Cookies aren't shared during InPrivate

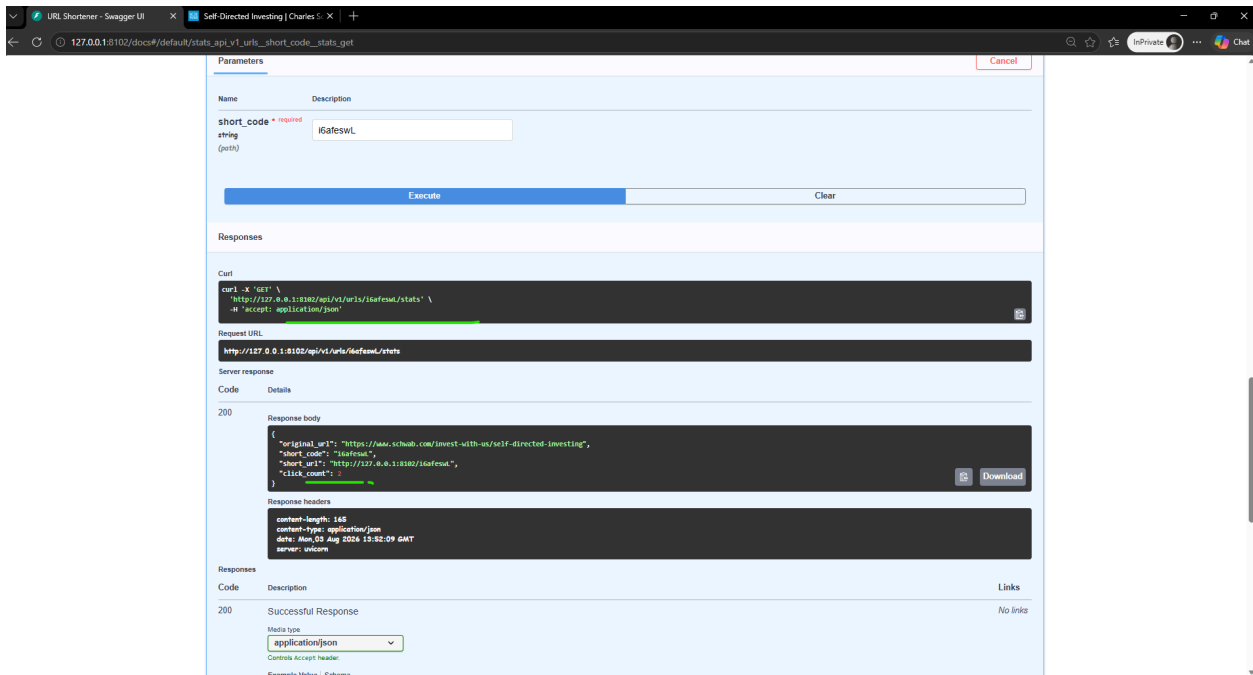
InPrivate browsing keeps cookies private to help prevent sites from sharing your data across the web. If a site doesn't work, you can temporarily allow cookie access.

More details

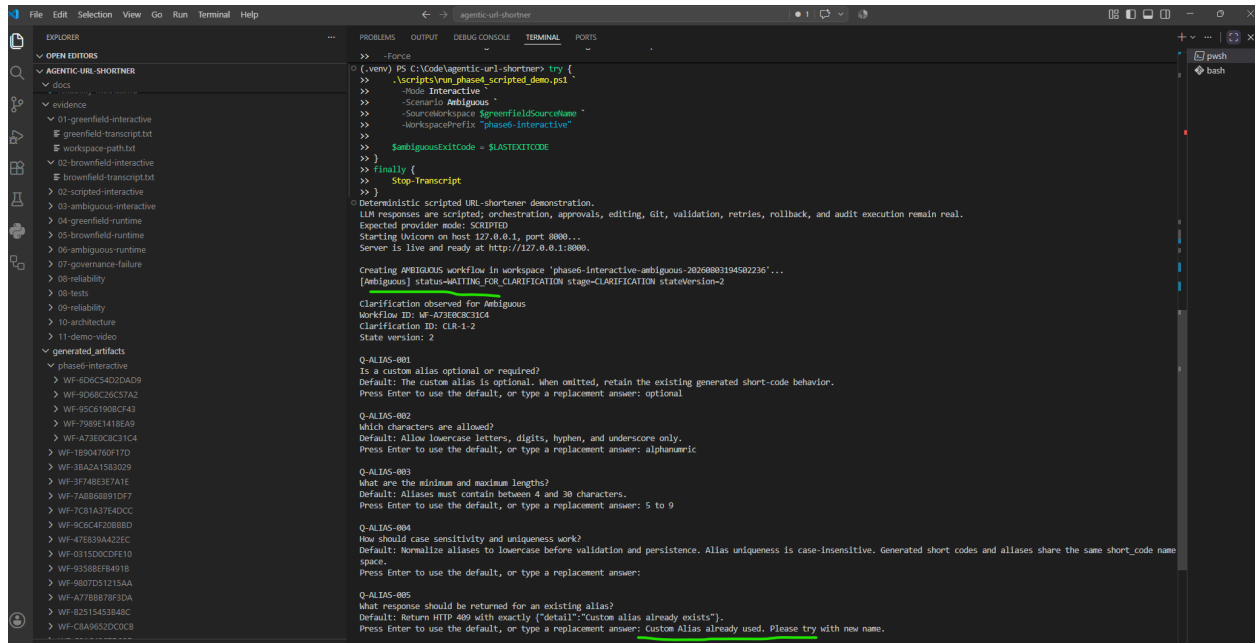




Clicked 2 times and stats are shown below



### 3. Ambiguous scenario



```
File Edit Selection View Go Run Terminal Help
agentic-url-shortener

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
>> -Force
(.venv) PS C:\code\agentic-url-shortener> try {
>> .\scripts\run_phase_scripted_demo.ps1
>> -Mode Interactive
>> -Scenario 'Ambiguous'
>> -SourceWorkspace 'greenfieldSourceVolume'
>> -WorkspacePrefix 'phases-interactive'
>> $ambiguousExitCode = $LASTEXITCODE
>> } finally {
>> Stop-Transcript
>> }
Deterministic scripted URL-shortener demonstration.
LHM responses are scripted; orchestration, approvals, editing, Git, validation, retries, rollback, and audit execution remain real.
Expected provider mode: SCRIPTED
Starting Uvicorn on host 127.0.0.1, port 8000...
Server is live and ready at http://127.0.0.1:8000.

Creating AMBIGUOUS workflow in workspace 'phases-interactive-ambiguous-20260803194502236'...
[ambiguous] status-WAITING_FOR_CLARIFICATION stage-CLARIFICATION stateVersion=2

Clarification observed for Ambiguous
Workflow ID: WF-A73E0C8C31C4
Clarification ID: CLR-1-2
State version: 2

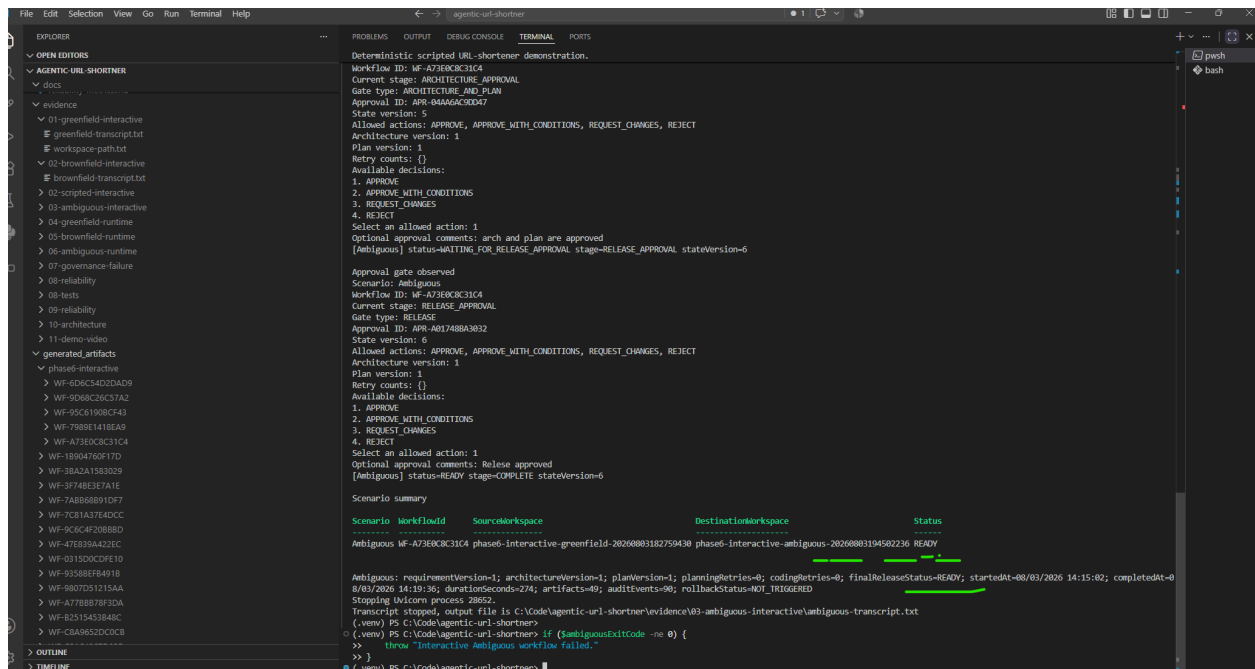
Q-ALIAS-001
Is a custom alias optional or required?
Default: The custom alias is optional. When omitted, retain the existing generated short-code behavior.
Press Enter to use the default, or type a replacement answer: optional

Q-ALIAS-002
Which characters are allowed?
Default: Allow lowercase letters, digits, hyphen, and underscore only.
Press Enter to use the default, or type a replacement answer: alphanumeric

Q-ALIAS-003
What are the minimum and maximum lengths?
Default: Aliases must contain between 4 and 30 characters.
Press Enter to use the default, or type a replacement answer: 5 to 9

Q-ALIAS-004
How should case sensitivity and uniqueness work?
Default: Normalize aliases to lowercase before validation and persistence. Alias uniqueness is case-insensitive. Generated short codes and aliases share the same short_code name space.
Press Enter to use the default, or type a replacement answer:

Q-ALIAS-005
What response should be returned for an existing alias?
Default: Return HTTP 400 with exactly ["detail": "Custom alias already exists"].
Press Enter to use the default, or type a replacement answer: Custom Alias already used. Please try new name.
```



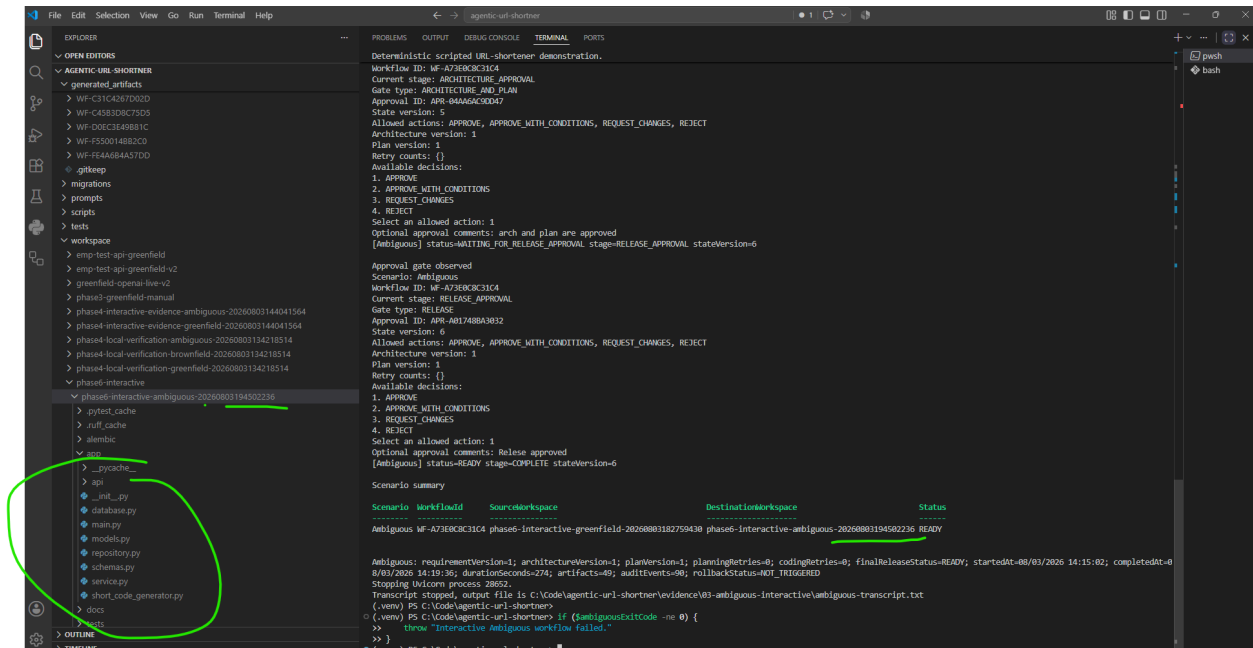
```
File Edit Selection View Go Run Terminal Help
agentic-url-shortener

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Deterministic scripted URL-shortener demonstration.
Workflow ID: WF-A73E0C8C31C4
Current stage: ARCHITECTURE_APPROVAL
Gate type: ARCHITECTURE_AND_PLAN
Approval ID: APR-906C26C37A2
State version: 5
Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
Architecture version: 1
Plan version: 1
Retry counts: {}
Available decisions:
1. APPROVE
2. APPROVE_WITH_CONDITIONS
3. REQUEST_CHANGES
4. REJECT
Select an allowed action: 1
Optional approval comments: arch and plan are approved
[ambiguous] status-WAITING_FOR_RELEASE_APPROVAL stage-RELEASE_APPROVAL stateVersion=6

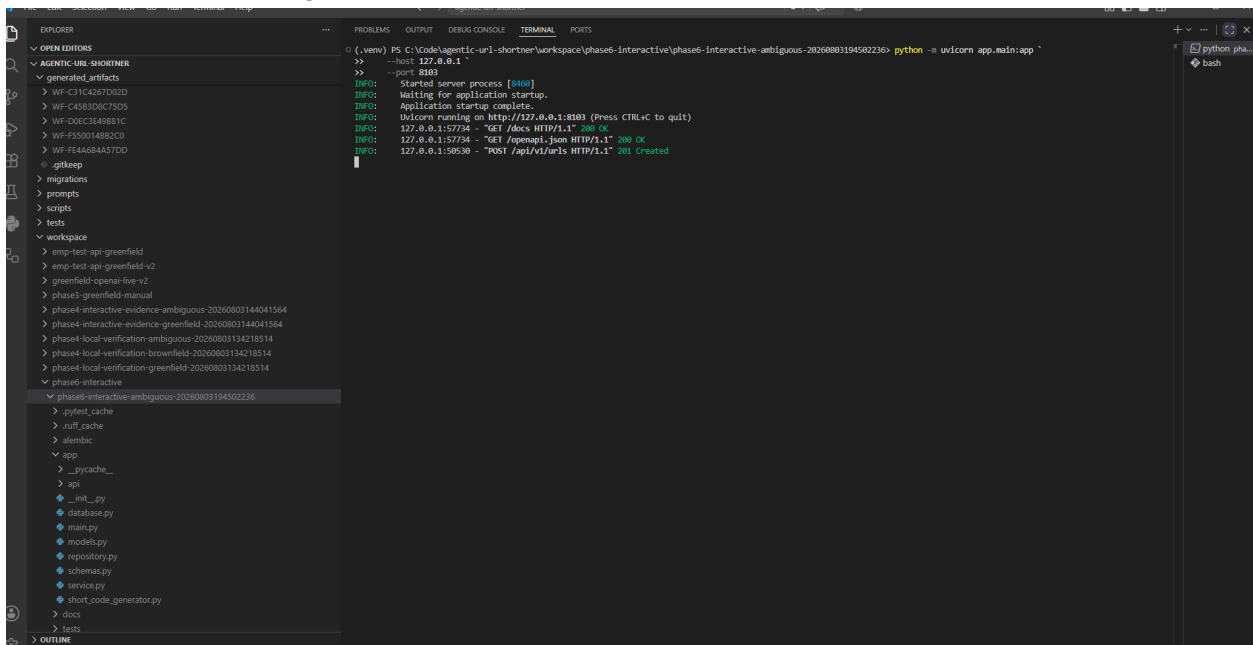
Approval gate observed
Scenario: Ambiguous
Workflow ID: WF-A73E0C8C31C4
Current stage: RELEASE_APPROVAL
Gate type: RELEASE
Approval ID: APR-907D51215AA
State version: 6
Allowed actions: APPROVE, APPROVE_WITH_CONDITIONS, REQUEST_CHANGES, REJECT
Architecture version: 1
Plan version: 1
Retry counts: {}
Available decisions:
1. APPROVE
2. APPROVE_WITH_CONDITIONS
3. REQUEST_CHANGES
4. REJECT
Select an allowed action: 1
Optional approval comments: Release approved
[ambiguous] status-READY stage-COMplete stateVersion=6

Scenario summary
Scenario WorkflowId SourceWorkspace DestinationWorkspace Status
-----
Ambiguous WF-A73E0C8C31C4 phases-interactive-greenfield-20260803182759430 phases-interactive-ambiguous-20260803194502236 READY

Ambiguous: requirementVersion=1; architectureVersion=1; planVersion=1; planningRetries=0; codingRetries=0; finalReleaseStatus=READY; startedAt=08/03/2026 14:15:02; completedAt=08/03/2026 14:19:36; durationSeconds=274; artifacts=0; auditEvents=0; rollbackStatus=NOT_TRIGGERED
Stopping Uvicorn process 28622.
Transcript stopped, output file is C:\code\agentic-url-shortener\workspace\greenfield\ambiguous-interactive\ambiguous-transcript.txt
(.venv) PS C:\code\agentic-url-shortener>
(.venv) PS C:\code\agentic-url-shortener> if ($ambiguousExitCode -ne 0) {
>> throw "Interactive Ambiguous workflow failed."
>> }
>>
(.venv) PS C:\code\agentic-url-shortener>
```



## Generated code running



URL Shortener - Swagger UI

127.0.0.1:8103/docs#/default/create\_url\_api\_v1\_urls\_post

Private

Chat

ExecuteClear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8103/api/v1/urls' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "original_url": "https://www.schub.com/invest-with-us/self-directed-investing",
    "custom_alias": "schub"
  }'
```

Request URL

http://127.0.0.1:8103/api/v1/urls

Server response

CodeDetails

201

Response body

```
{
  "original_url": "https://www.schub.com/invest-with-us/self-directed-investing",
  "short_code": "schub",
  "short_url": "http://127.0.0.1:8103/schub"
}
```

Download

Response headers

```
content-length: 146
content-type: application/json
date: Mon, 05 Aug 2024 14:23:36 GMT
server: envoy
```

Responses

| Code | Description         | Links    |
|------|---------------------|----------|
| 201  | Successful Response | No links |